Transmission electron microscopy (TEM)

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1 Introduction

Transmission electron microscopy (TEM) is a tool to investigate materials on the smallest scales in the nanometer range. To achieve this, it does not use light as its probe like in optical microscopy but an electron beam. This electron beam passes through the sample and by studying how it interacted with the sample an image can be formed about the structure of the investigated material. In this experiment we will use two devices, A Philips CM30 TEM to study a silicon crystal and the Thermo Fisher Scientific Spectra 200 to study e PbS nanoparticles. Several TEM techniques will be used. These include Selected-area electron diffraction (SAED), bright-field and dark-field TEM imaging (BF-TEM and DF-TEM). In order to understand how TEMs work, we must first understand the path that the electron beam takes through the device. Afterwards we must understand the crystal structure of materials and how electron diffraction off these crystals works.

1.1 TEM device

In Figure 1 the electron ray path through the TEM is shown for the two setups, to investigate the diffraction pattern or form an image. The general structure is as follows. The electron beam is released from an electron gun and accelerated up to velocities above 80% the speed of light. The beam passes through a set of condenser lenses, where high-angle electron are removed from the beam and then the sample. The electrons interact with the sample and a certain fraction is transmitted. Two types of images can be made on the viewing screen, depending on the lens setup behind the sample. The electron beam is guided through the device by electromagnetic lenses.

Image mode:
There are two types of images, the Bright-Field (BF) and the Dark Field (DF). In the image mode an objective aperture is placed at the back focal plane of the objective lens after the sample. If the zero-beam is selected a bright-field image is formed, if a diffracted beam is selected a dark-field image is formed [1]. What arrives at the viewing screen, a fluorescence screen, is an image of the image plane of the objective lens. For the BF-image this is shown in Figure 2a, for a DF-image in Figure 2b. The lenses used in TEMs are electromagnets.

Diffraction mode:
In case of the Selected Area Diffraction mode, the objective aperture is removed and an intermediate aperture is inserted. This produces an image of the back focal plane of

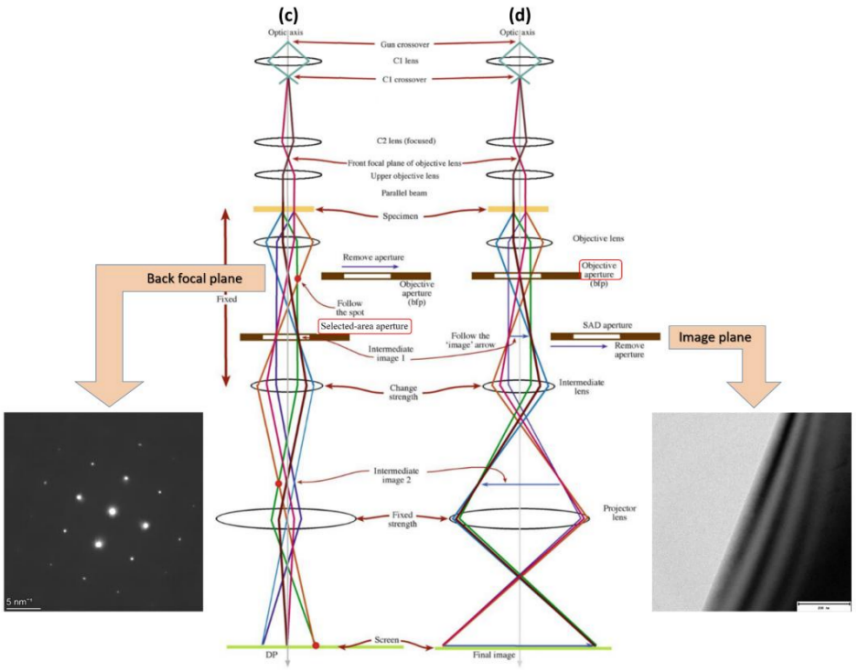
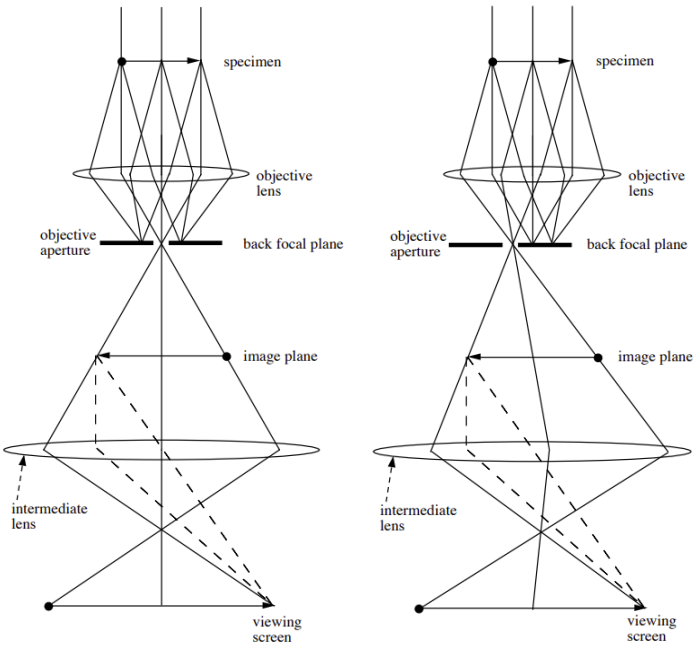


Figure 1: TEM electron path in the diffraction mode and the image mode [2].

the objective lens [1]. The electron beam path in the SAED mode can be seen in Figure 3.



(a) bright-field image setup (b) dark-field image setup

Figure 2: Electron ray path for the two image modes, BF- and DF-images [1].

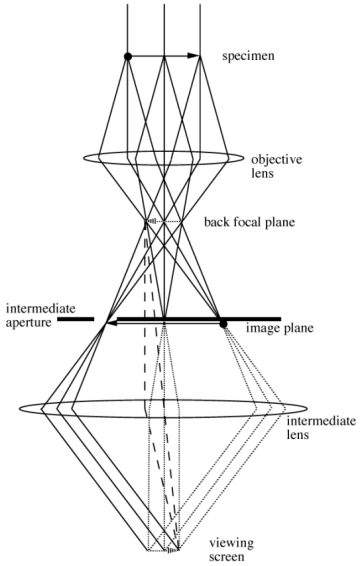


Figure 3: Electron ray path for the SAED mode [1].

The switch from image mode to diffraction mode and vice versa can be done by changing the excitation of the intermediate lens which then either focuses the image plane (BF-DF-image) or the back focal plane (diffraction). In practice this only requires a button press. Next we describe how the electrons interact with the sample.

1.2 Electron diffraction

Electrons can be described as wave-objects in quantum theory and have a wavelength according to the de Broglie relation $\lambda = h/p$. For the interaction with the sample both the properties of the electron and the structure of the sample is important. For this we describe what real and reciprocal lattices are. The following calculations will be done In the case of kinematical diffraction, where an electron only interacts once in the sample,

1.2.1 Real and reciprocal lattice

In the fields of physics and materials science, particularly in crystallography, real space and reciprocal space serve as two different perspectives on the same object.

- **Real Space**: This refers to the three-dimensional world, where things like atoms and molecules exist. In the context of crystallography, real space is used to describe the precise arrangement of atoms within a crystal lattice, capturing their physical positions in space.
- **Reciprocal Space**: In contrast, reciprocal space is a mathematical construct where points correspond to spatial frequencies or periodicities, rather than physical locations. Reciprocal space shows periodic patterns and symmetries inherent in a crystal lattice.

1.2.2 Bragg's Condition

X-ray waves or electron beam waves serve as the incident rays in diffraction experiments and then interact with the sample. A diffraction maximum occurs only when these waves

interfere constructively, meaning their phase differences are whole multiples of 2π. This condition corresponds to a path difference that is a multiple of the wavelength, expressed as $n\lambda$, where n is an integer and λ is the wavelength.

In practical terms, the path difference for constructive interference in a crystal lattice is described by Bragg's law. The standard mathematical form of Bragg's law is:

$$2d \sin \theta_{\mathrm{B}} = n\lambda \ , \tag{1}$$

where d is the spacing between lattice planes and θ_{B} is the angle of incidence. This relationship governs the conditions for diffraction maxima in X-ray-and electron crystallography.

1.2.3 Laue condition and Ewald sphere

Bragg's law as written in Equation 1, describes the Bragg condition in real space. If we consider $n = 1$ from now on, it can also be described in reciprocal space. This is known as the Laue condition. It can be expressed as:

$$\vec{\mathbf{k}} - \vec{\mathbf{k_0}} = \vec{\mathbf{g}} \ , \tag{2}$$

where $\vec{\mathbf{k}}$ is the wavevector of the scattered electron beam, $\vec{\mathbf{k_0}}$ of the incident beam and $\vec{\mathbf{g}}$ is a reciprocal lattice vector. $\vec{\mathbf{g}}$ is typically written in terms of Miller indices (hkl). This shows us that in reciprocal space constructive interference happens, if the difference in wavevectors of a scattered beam to its initial wavevector is equal to a reciprocal lattice vector, which relates to periodicities in the real space lattice. This condition can be geometrically visualized with the so-called Ewald sphere construction in reciprocal space. Here the center of the sphere is at the starting point of $\vec{\mathbf{k_0}}$ and ends where $\vec{\mathbf{k_0}}$ meets the origin of the reciprocal lattice. The length of these vectors is $|\vec{\mathbf{k}}| = \frac{1}{\lambda}$ and $|\vec{\mathbf{g}}| = \frac{1}{d}$. The angle between $\vec{\mathbf{k_0}}$ and $\vec{\mathbf{k}}$ is twice the Bragg angle θ_{B} [2]. A diagram of what the Bragg condition looks like in real space and reciprocal space is shown in Figure 4.

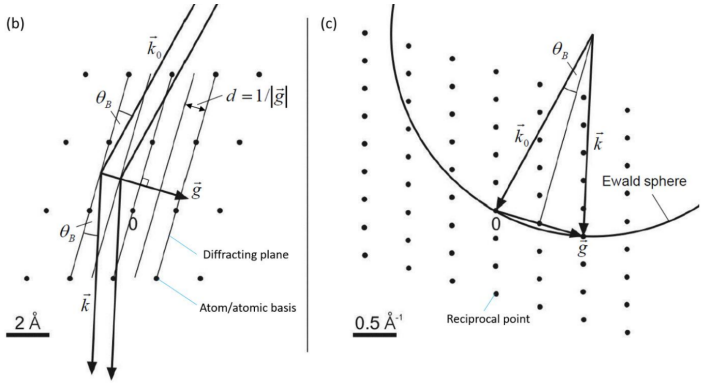


Figure 4: b) Diagram of the Bragg condition in real space. c) Diagram of Laue condition and Ewald sphere in reciprocal space.

1.2.4 Zone Axis

For highly accelerated electrons, as is the case for TEM, their wavelength is smaller than the lattice spacing of typical crystals. This means that the Bragg angle is very small, on

the order of a few milliradians. From this follows, that lattice planes that are hit edge-on by the electron beam already almost satisfy the Bragg condition. If multiple diffraction planes have one common direction $\vec{\mathbf{R}}$, the collection of these diffraction planes is called a zone, with the axis parallel to $\vec{\mathbf{R}}$ being called the zone axis. This is commonly expressed as [uvw], instead of writing the 3D-vector $\vec{\mathbf{R}}$. These vector components represent the smallest ratio of integers of the $\vec{\mathbf{R}}$-vector [2]. Figure 5 shows a diagram of a zone axis $\vec{\mathbf{R}}$ and different diffraction planes within the zone. As an example if the two reflections are $\vec{\mathbf{g_1}} = (200)$ and $\vec{\mathbf{g_2}} = (220)$, then the zone axis is calculated from the vector product $\vec{\mathbf{g_1}} \times \vec{\mathbf{g_2}} = (004)$, where the zone axis is now [u v w] = [0 0 1].

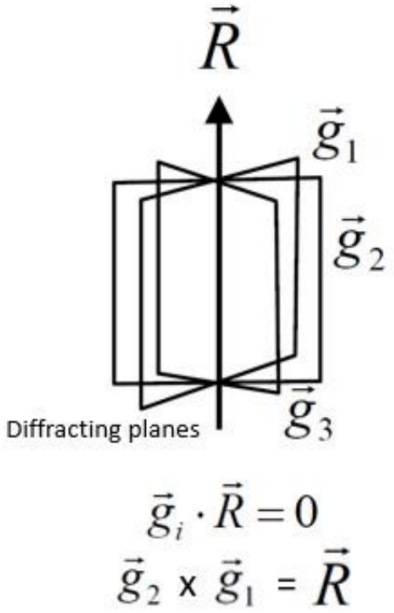


Figure 5: Diagram of zone axis $\vec{\mathbf{R}}$ and diffraction planes $\vec{\mathbf{g}}$ [2].

1.2.5 Excitation error

The intensity of the diffracted beam I_g in the kinematical approximation has two components, the lattice amplitude $G(\vec{\mathbf{g}})$ and the structure amplitude $F(\vec{\mathbf{g}})$. The lattice amplitude depends on the shape of the crystal.
If one considers an infinite crystal, $G(\vec{\mathbf{g}})$ contains δ-functions at each reflection $\vec{\mathbf{g}}$ in the reciprocal lattice. Constructive interference occurs, when the Ewald sphere intersects reciprocal lattice points as described by the Laue condition. This then results in an infinite intensity of the diffracted beam, which is however unrealistic as electrons can only penetrate thin samples, for the kinematical approximation to be valid. This is rectified in a dynamical theory.
In a more realistic model of a sample with some thickness t, the resulting reciprocal lattice points are stretched in direction of the sample normal direction. This represents a lattice amplitude $G(\vec{\mathbf{g}})$ with intensities of a sinc-function along the sample normal for each reciprocal lattice "rod".

$$G(\vec{\mathbf{g}}) \propto \frac{\sin \pi t s_\mathrm{g}}{\pi t s_\mathrm{g}} \cdot t \ , \tag{3}$$

where s_g is the so-called excitation error. The meaning of it, is that as the Ewald sphere is moved away from a $\vec{\mathbf{g}}$ point the intensity of the diffracted beam does not vanish immediately but decreases the further away it is, because the "rod" still intersects the Ewald sphere [2]. This is illustrated in Figure 6.

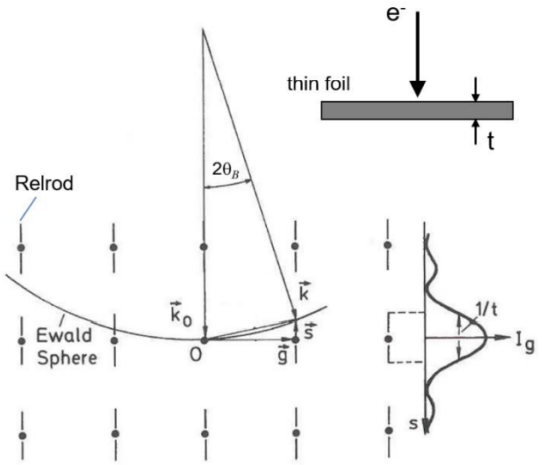


Figure 6: This diagram shows how the Ewald sphere intersects with the elongated reciprocal lattice "rods" in the case of a sample with thickness t and the resulting G intensities [2].

1.2.6 Structure amplitude

To analyze the structure amplitude $F(\vec{\mathbf{g}})$, we must consider the structure of the unit cell of the sample crystal. If there are multiple atoms in the unit cell the diffracted beams could interfere destructively and result in $F(\vec{\mathbf{g}}) = 0$ for some reflections $\vec{\mathbf{g}}$. These are the "kinematically forbidden reflections". In general $F(\vec{\mathbf{g}})$ is:

$$F(\vec{\mathbf{g}}) = \sum_{j}^{N} = f_j \exp(-2\pi i \vec{\mathbf{g}} \cdot \vec{\mathbf{r_j}}) \ , \tag{4}$$

where f_j are the atomic form factors and r_j are the position vectors of atom j in the unit cell [2]. Here we calculate what reflections are kinematically forbidden in the case of a face-centered-cubic (FCC) and diamond lattice.

FCC-lattice:

There are four unique positions in the unit cell $\vec{\mathbf{r_1}} = (0,0,0)$, $\vec{\mathbf{r_2}} = (0,1/2,1/2)$, $\vec{\mathbf{r_3}} = (1/2,0,1/2)$ and $\vec{\mathbf{r_4}} = (1/2,1/2,0)$. This leads to:

$$F_{\mathrm{FCC}} = f_{\mathrm{A}} \left[1 + \exp\left(-\pi i(k+l)\right) + \exp\left(-\pi i(h+l)\right) + \exp\left(-\pi i(h+k)\right)\right] \tag{5}$$

When Miller indices have mixed parity, some are odd some are even, then the structure amplitude is zero. The allowed reflections are those where all Miller indices are either odd or even [2]. This is summarized as:

$$F_{\mathrm{FCC}} = \begin{cases} 0 & \text{h,k,l: mixed parity} \\ 16|f_{\mathrm{A}}|^2 N & \text{h,k,l: unmixed parity} \end{cases}$$

This is illustrated in Figure 7.

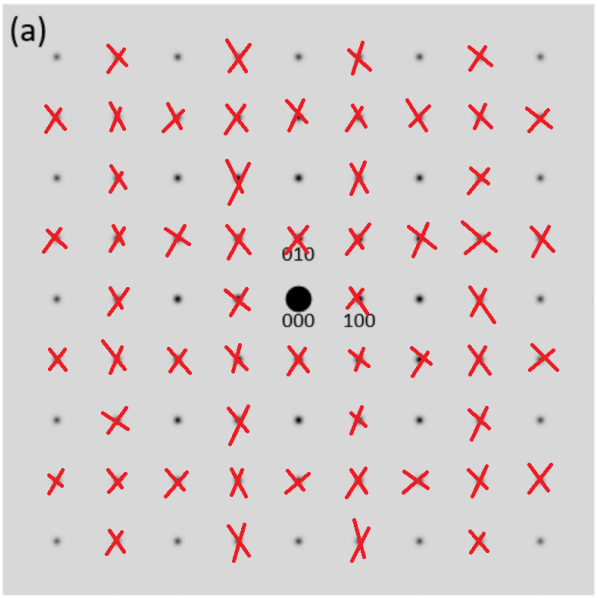


Figure 7: Simulated of [001] SAED pattern for the FCC-lattice. Indicated in red crosses are the forbidden reflections.

Diamond-lattice:
There are eight unique positions inside the unit cell $\vec{\mathbf{r_1}} = (0,0,0)$, $\vec{\mathbf{r_2}} = (0,1/2,1/2)$, $\vec{\mathbf{r_3}} = (1/2,0,1/2)$, $\vec{\mathbf{r_4}} = (1/2,1/2,0)$, $\vec{\mathbf{r_5}} = (1/4,1/4,1/4)$, $\vec{\mathbf{r_6}} = (3/4,3/4,1/4)$, $\vec{\mathbf{r_7}} = (3/4,1/4,3/4)$ and $\vec{\mathbf{r_8}} = (1/4,3/4,3/4)$. This leads to:

$$F_{\text{diamond}} = F_{\text{FCC}} \cdot [1 + \exp(-\pi i(h+k+l)/2)] \tag{6}$$

The structure amplitude is zero if some of h,k,l are mixed parity. It also becomes zero if all h,k,l are either even or odd with $h+k+l = 4n+2$ [2]. This is summarized as:

$$F_{\text{diamond}} = \begin{cases} 0 & \text{h,k,l: mixed, unmixed, h+k+l = 4n + 2} \\ 64|f_{\text{A}}|^2 N & \text{h,k,l: unmixed,h+k+l=4n} \\ 32|f_{\text{A}}|^2 N & \text{h,k,l: unmixed,h+k+l=4n+1} \end{cases}$$

This is illustrated in Figure 8.

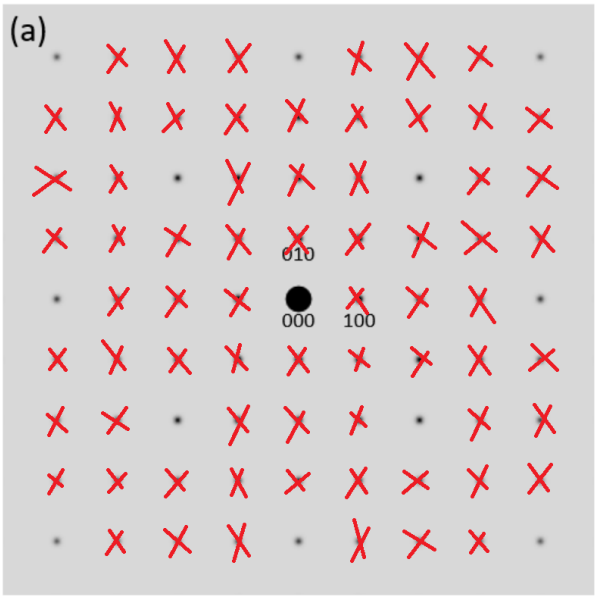


Figure 8: Simulated [001] SAED pattern for the diamond-lattice. Indicated in red crosses are the forbidden reflections.

1.2.7 Kikuchi-Lines

So far we have considered that the electron only interacts once in the crystal. When an electron interacts multiple times in the crystal, this is called dynamical diffraction. Inelastic scattering can occur in the crystal, when the electron loses energy by exciting a lattice vibration. Kikuchi-lines are formed, when this scattered electron fulfills the Bragg condition. These Kikuchi-lines are line-shaped Bragg reflections which show up as lines of excess or deficiency intensities in the diffraction pattern. Their origin is in the angular dependence of the atomic form factor. Some electrons are scattered towards the side K_{hkl} whereas most are scattered in forward-direction $K_{\overline{h},\overline{k},\overline{l}}$. These two beams satisfy the Bragg condition and have an angle of $2\theta_{\mathrm{B}}$ between them. Some of them can Bragg diffract into the path of the other and due to their different amplitudes, directions of smaller and higher intensities occur. Due to the existence of multiple pairs of these diffracted beams, these excess and deficiency regions appear as lines in the diffraction pattern [2]. Figure 9 shows the location of the Kikuchi-lines in a diamond lattice.
If you wanted to move the Kikuchi-lines away from their Bragg reflections (sharp spots), you could tilt the sample. This changes the angle at which the electron beam interacts with the crystal planes. This leads to a shifting of the Kikuchi lines relative to the Bragg reflections. Note that the diffraction spots do not move. With this technique one can achieve the two-beam condition, where only the zero-beam and the diffraction beam of one reflection $\vec{\mathbf{g}}$ are excited. The crystal can be tilted until the excess line covers the reflection spot $\vec{\mathbf{g}}$ and the deficit line covers the zero-beam [2].

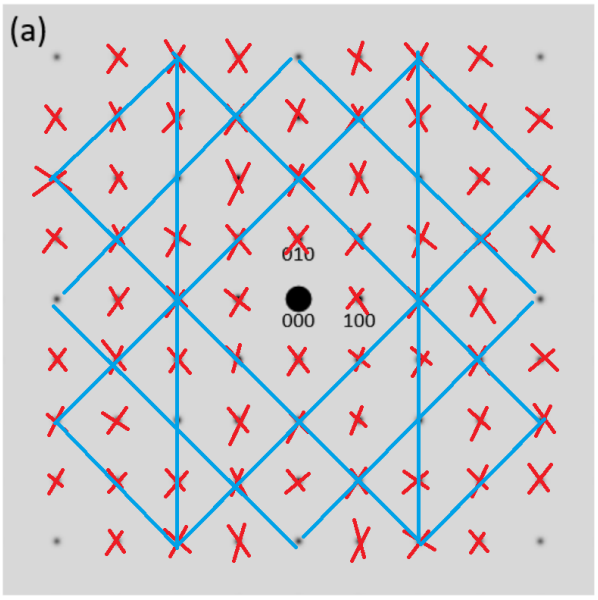


Figure 9: Simulated SAED pattern for the diamond-lattice. Indicated in red crosses are the forbidden reflections. Indicated in blue are the Kikuchi-lines.

1.2.8 Extinction distance

In the dynamical theory the electron can perform multiple interactions in the sample. If the electron is diffracted multiple times within a distance z in the sample in Bragg condition, the intensity of the diffracted beam I_g is shifted to the zero-beam I_0 and vice versa for increasing distances [2]. The intensity of the diffracted beam is given by

$$I_g = 1 - I_0 \propto \frac{\sin^2(\pi t s_{\text{eff}})}{(\pi t s_{\text{eff}})^2} \cdot t^2 \ , \tag{7}$$

where $s_{\text{eff}} = \sqrt{s_g^2 + \xi_g^{-2}}$ and ξ_g is the so-called extinction distance. Due to this modulation of the intensity, this is called the "Pendellösung effect". The meaning of ξ_g is the periodicity of the I_g-I_0-relation. Over a distance of $\xi_g/2$ I_0 fully turns into I_g and at one ξ_g the I_0 and I_g are back at the starting position [2].

1.3 Imaging effects

Thickness fringes:
If the TEM is set in the imaging mode and one images a wedge-shaped sample in the DF-TEM configuration with the two-beam condition $s_g = 0$, one will observe thickness fringes in the DF-image due to the fulfilled Bragg condition. These are alternating bright and dark stripes. If one tilts the sample away from the Bragg condition as the excitation error s_g becomes larger, the spacing between the fringes becomes smaller. The fringes appearing at $s_g = 0$ is due to a dynamical effect and not predicted in the kinematical approximation. If $s_g >> 0$ the intensities predicted by the kinematical and the dynamical theory become the same [2].
Bend contours:
If the crystal is bent, the lattice is not homogeneous anymore. The orientation of diffraction planes changes from one place to another. This in turn changes the excitation error

for a reflection at each place, which leads to fringes appearing in the image. Because these fringes are due to a bending of the sample, they are also called bend contours.

References

[1] Brent Fultz and James M Howe. *Transmission electron microscopy and diffractometry of materials*. Springer Science & Business Media, 2012.

[2] *M43 Transmission Electron Microscopy (TEM) lab manual*. 2025.